Maria Paola Vaccaro

Born on July 11th, 1997
in Larino (CB), Italy
Currently in Heidelberg, Germany

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Scientific interests

- Computational astrophysics
- Stellar evolution
- Gravitational waves sources
- Black holes dynamics
- Stellar clusters

Numerical tools

Python	●●●●
C/C++	●●●○
Matlab	●●●○

Operating systems

Linux/Unix	●●●○
Windows	●●●○

Languages

Italian (mother tongue)
English (fluent, level C1)
Spanish (basic, level A2)
German (basic, level A1)

Education

Sep. 2023 – 2026 **PhD in Astronomy and Cosmic Physics**
At Institute for Theoretical Astrophysics (ITA), ZAH, Heidelberg University in Heidelberg, Germany.
Supervisor: Prof. Michela Mapelli - Thesis title: "Binary Black Hole population in the Local Universe"

Oct. 2020 – Sep. 2022 **Master degree in Astrophysics**
At Università degli Studi di Padova in Padua (PD), Italy˙ Final grade: 110/110 *cum laude*, GPA: 29.4/30
Supervisor: Prof. Michela Mapelli - Thesis title: "Hierarchical Black Holes Mergers in Active Galactic Nuclei"

Sep. 2019 – Jul. 2020 **Master course in Matter Physics**
At Università di Pisa in Pisa (PI), Italy - GPA: 28.8/30

Sep. 2016 – Sep. 2019 **Bachelor degree in Physics**
At Università di Pisa in Pisa (PI), Italy.
Final grade: 110/110 *cum laude*, GPA: 28.6/30
Supervisor: Prof. Pier Giorgio Prada Moroni - Thesis title: "Cooling evolution of white dwarfs and cosmochronology"

Sep. 2011 – Jul. 2016 **High school diploma**
At Liceo Scientifico Alfano da Termoli in Termoli (CB), Italy.
Final grade: 100/100 *cum laude*

Aug. 2014 – Jul. 2015 **High school exchange year**
At Mead High School in Mead, Colorado, USA.
GPA: 4.0/4 - ACT score: 33/36

Funding

Member of the Young Research Convent of the Heidelberg Cluster of Excellence STRUCTURES since October 1st, 2024. Program funded by the German Excellence Strategy fund.

Pre-doctoral fellowship February 1st – August 31st, 2023
DEMOBLACK ERC Team at Università degli Studi di Padova, Padua (PD), Italy. Supervisor: Prof. Michela Mapelli. Research project: "Hierarchical evolution of stellar-sized binary black holes in active galactic nuclei".

Internship in GW group October 1st – December 31st 2022
IFAE-UAB, Barcelona, Spain. Supervisors: Dr. Lluïsa-Maria Mir i Martínez and Prof. Mario Martínez Pérez

Current activity

I am a computational astrophysicist, working in the field of gravitational-wave astronomy and black hole dynamics. I am the main developer of a semi-analytical code for the active galactic nucleus (AGN) disk scenario for hierarchical black hole mergers.

Conferences and workshop

Contributed Talk — **University of Tokyo, Tokyo, Japan**
RESCEU-NBIA workshop on GW sources — **December 7th – 15th, 2023**

Talk title: "The impact of Gas Hardening on Hierarchical Black Hole Mergers in Active Galactic Nuclei Disks". Contribution to the **discussion panel** "Many-Body Systems".

Flash Talk — **University of Milano Bicocca, Milan, Italy**
Workshop "GwPopNext" — **July 10th – 14th, 2023**

Talk title: "Semi-analytical Model for Hierarchical Binary Black Hole Mergers".

Invited poster presentation — **Lorentz Center, Leiden, Netherlands**
Workshop "The Renaissance of Stellar Black-Hole Detections" — **June 26th – 30th, 2023**

Invited Speaker — **Venetian institute of science, literature and art, Venice, Italy**
Physics of Data Workshop — **April 19th – 21st, 2023**

Talk title: "How do black holes grow? Active galactic nuclei as a breeding ground for intermediate-mass black holes"

Scientific publications

As first author

Impact of gas hardening on the population properties of hierarchical black hole mergers in AGN disks
M. Paola Vaccaro, Michela Mapelli, Carole Périgois, Dario Barone, M. Celeste Artale, Marco Dall'Amico, Giuliano Iorio, Stefano Torniamenti, 2023, *Accepted for publication on A&A*, arXiv:2311.18548

Others

Hierarchical binary black hole mergers in globular clusters: mass function and evolution with redshift
Stefano Torniamenti, Michela Mapelli, Carole Périgois, Manuel Arca Sedda, M. Celeste Artale, Marco Dall'Amico, M. Paola Vaccaro, 2024, *Submitted to A&A*, arXiv:2401.14837

GW190521: a binary black hole merger inside an active galactic nucleus?
Sophia Morton, Stefano Rinaldi, Alejandro Torres-Orjuela, Andrea Derdzinski, M. Paola Vaccaro, Walter Del Pozzo, 2023, Phys. Rev. D, 108, 123039

The hitchhiker's guide to the galaxy catalog approach for gravitational wave cosmology
Jonathan R. Gair, Archisman Ghosh, Rachel Gray, ..., Maria Paola Vaccaro et al., 2022, AJ, 166, 22

Courses and schools

- 4th G2Net Training School "A network for Gravitational Waves, Geophysics and Machine Learning" hosted by Aristotle University of Thessaloniki (remote attendance) — March 28th-30th, 2023
- Python Course (level advanced) hosted by ICFO in Barcelona, Spain — November 24th-30th, 2022
- GW Open Data Workshop hosted by GWOSC (online event) — May 23rd-25th, 2022

- IPAM Workshop III: Source inference and parameter estimation in Gravitational Wave Astronomy (remote attendance) November 15th-19th, 2021

Outreach

- Staff member of the cultural association Scholé based in Roccella Jonica (RC), Italy since 2020
- Member of the organizing committee for the Scholé “School of Physics” for high schoolers since 2022
- Speaker and staff member at the 2023 Scholé Summer School titled "PHYSIS" ("Nature"). Talks:
  - "Understanding Nature with Quantum Mechanics" with Prof. Giancarlo Cella (Pisa, Italy) July 25th, 2023
  - "The concept of space from Newton to Einstein" with Prof. Paolo Bussotti (Udine, Italy) July 29th, 2023

Awards and other activities

- Don Peppe Presutto award from Circolo CFR Serracapriola for academic achievements 2016, 2020, 2023
- Aupair program in Cork, Ireland for language enhancement 2018
- Lions Club Tifernus award and 1000€ scholarship for educational achievements 2016
- Internship in the local newspaper Termolionline in Termoli (CB), Italy 2014